Both Closing Problems in `arXiv:1301.5799`: Literature Status

`fa_banach_001`, `agent_lane_12`

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Status

Literature already answered. The two closing questions in Cuth's *Noncommutative Valdivia compacta* are answered in separate later papers. The first answer identifies the requested function-space property, with a countable-compactness condition in the merely dense formulation; the second gives a four-way equivalence stronger than the three requested implications.

Source: Problem 1

On PDF page 13 of arXiv:1301.5799, Problem 1 asks for a topological property (T) of $(C(K), \tau_p(D))$ which characterizes those dense (respectively, dense and countably closed) sets $D \subset K$ induced by a retractional skeleton.

Cuth–Kalenda, *Monotone retractability and retractional skeletons*, arXiv:1403.4480, explicitly says that its Theorem 1.4 answers this problem. That theorem (PDF page 2; proof on page 11) states that, for compact K and dense $D \subset K$, the following are equivalent:

(i) D is induced by a retractional skeleton in K;

(ii) D is countably compact and $(C(K), \tau_p(D))$ is monotonically Sokolov.

Thus the sought function-space property is *monotone Sokolovness*. In the source's dense-and-countably-closed branch, D is automatically countably compact. For a merely dense D, the exact characterization keeps countable compactness as an additional condition on D.

Source: Question 1

The same source page asks whether, for compact K, the following implications hold among:

(i) $C(K)$ has a 1-projectional skeleton;

(ii) a convex symmetric set is induced by a retractional skeleton in $(B_{C(K)^*}, w^*)$;

(iii) a convex symmetric set is induced by a retractional skeleton in $P(K)$.

Specifically, it asks for (iii)⇒(ii), (ii)⇒(i), and (iii)⇒(i).

Cuth, *Simultaneous projectional skeletons*, arXiv:1305.1438, states that Theorem 4.1 answers this question. The theorem (PDF page 6) proves the equivalence of:

(i) $C(K)$ has a 1-projectional skeleton;

(ii) there is a convex symmetric set induced by a retractional skeleton in $(B_{C(K)^*}, w^*)$;

(iii) there is a convex set induced by such a skeleton in the same dual ball;

(iv) there is a convex set induced by a retractional skeleton in $P(K)$.

Condition (iv) is weaker than the source's symmetric $P(K)$ condition. Therefore all three implications asked in Question 1 have affirmative answers.

Scope

This is a literature-status identification, not a new proof. Both answering papers explicitly cite and answer the numbered problem or question in arXiv:1301.5799. The dense-only formulation above is reported with the exact countable-compactness qualification of Theorem 1.4.

References

[1] M. Cuth, *Noncommutative Valdivia compacta*, Comment. Math. Univ. Carolin. 55 (2014), no. 1, 53–72; arXiv:1301.5799.

[2] M. Cuth, *Simultaneous projectional skeletons*, J. Math. Anal. Appl. 411 (2014), 19–29; arXiv:1305.1438.

[3] M. Cuth and O. F. K. Kalenda, *Monotone retractability and retractional skeletons*, J. Math. Anal. Appl. 423 (2015), 18–31; arXiv:1403.4480.